

Supplementary Material for MolBridge: Atom-Level Joint Graph Refinement for Robust Drug-Drug Interaction Event Prediction

Anonymous Submission

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1 Atom Feature Encoding Details

This section provides the complete atom feature schema used for molecular graph representation in MolBridge. Each atom in a molecule is encoded as a 75-dimensional feature vector composed of the following descriptors:

Table 1: Detailed Atom Feature Encoding Scheme

Feature	Dim.	Description
Atomic symbol	44	One-hot encoding over [C, N, O, S, F, Si, P, Cl, Br, Mg, Na, Ca, Fe, As, Al, I, B, V, K, Tl, Yb, Sb, Sn, Ag, Pd, Co, Se, Ti, Zn, H, Li, Ge, Cu, Au, Ni, Cd, In, Mn, Zr, Cr, Pt, Hg, Pb, Unknown]
Atomic degree	11	One-hot encoding over [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10] representing the degree of atoms in a drug molecule
Implicit valence	7	One-hot encoding over [0, 1, 2, 3, 4, 5, 6] representing the implicit valence of atoms
Formal charge	1	Integer scalar representing the formal charge of the atom (typically ranges from -3 to +3)
Radical electrons	1	Integer scalar representing the number of free radical electrons (typically ranges from 0 to 2)
Hybridization	5	One-hot encoding over [SP, SP2, SP3, SP3D, SP3D2] representing the atomic hybridization mode
Aromaticity	1	Binary indicator (0/1) denoting whether the atom is aromatic
Total hydrogens	5	One-hot encoding over [0, 1, 2, 3, 4] representing the total number of hydrogen atoms attached
Total	75	

These features are extracted using RDKit 2024.9.6 and are concatenated to form the initial node feature matrix $F \in \mathbb{R}^{N \times 75}$ for each molecule, where N is the number of atoms. This feature representation captures both topological and chemical properties essential for modeling drug-drug interactions.

2 Complete Implementation Details

This section provides comprehensive experimental configurations and baseline comparison details.

2.1 Baseline Methods and Sources

For transductive experiments on Deng’s and Ryu’s datasets:

- **Results from literature:** DeepDDI, R-GCN, GoGNN, TrimNet-DDI, SSI-DDI, MUFFIN, and MRCGNN results are directly obtained from [1].
- **Re-implemented methods:** DSN-DDI, CSSE-DDI, and TIGER are re-implemented based on their official configurations and open-source code.

For inductive generalization experiments on DrugBank:

- All baseline results and experimental protocols are obtained from [2], which provides a comprehensive evaluation framework for inductive DDI prediction under S1 (one seen drug, one unseen) and S2 (both drugs unseen) settings.

2.2 Experimental Setup

Data splitting:

- Following the protocol in MRCGNN [1], we perform five-fold cross-validation.
- Each fold is further divided into training, validation, and test sets at a 7:1:2 ratio.

Evaluation metrics:

- Accuracy (Acc.): The proportion of correctly predicted drug pairs.
- Macro-F1: The harmonic mean of macro-precision and macro-recall.
- Macro-Precision (Macro-Pre.): Average precision across all DDI event classes.
- Macro-Recall (Macro-Rec.): Average recall across all DDI event classes.
- All results are averaged across five folds with standard deviations reported.

2.3 Training Configuration

Hyperparameters:

- Batch size: 512
- Learning rate: 0.005
- Random seed: 42
- Number of GFormer layers: 3
- Optimizer: AdamW with weight decay 0.01
- Dropout rate: 0.1
- Maximum epochs: 500
- Early stopping: Based on validation accuracy and macro-F1 score

Model selection:

- Best-performing model is selected based on combined Accuracy and Macro-F1 performance on the validation set.
- Final reported results are obtained by averaging performance across all five folds.

2.4 Hardware and Software Environment**Hardware:**

- CPU: Intel Xeon Platinum-8457C
- GPU: NVIDIA L20 (48GB memory)
- RAM: 128GB

Software:

- Operating System: Ubuntu 22.04.1 LTS
- Python: 3.9
- PyTorch: 2.4.0 (with CUDA 11.8 support)
- RDKit: 2024.9.6
- Additional libraries: NumPy, Pandas, scikit-learn 1.6.1, PyYAML, tqdm

3 Complete Ablation Study Results

To systematically evaluate the contribution of each component in MolBridge, we conducted comprehensive ablation experiments on Deng’s dataset. We examined four key variants:

- **w/o SCM**: Replace the Structure Consistency Module with standard GCN layers (no residual connections or layer normalization).
- **w/o A'** : Remove the original chemical bond adjacency matrix, using only the attention-based reconstructed graph.
- **w/o A_r** : Remove the attention-based graph reconstruction module, using only explicit chemical bonds.
- **w/o Joint**: Remove the joint molecular graph construction strategy; model drugs independently and fuse their representations afterward.

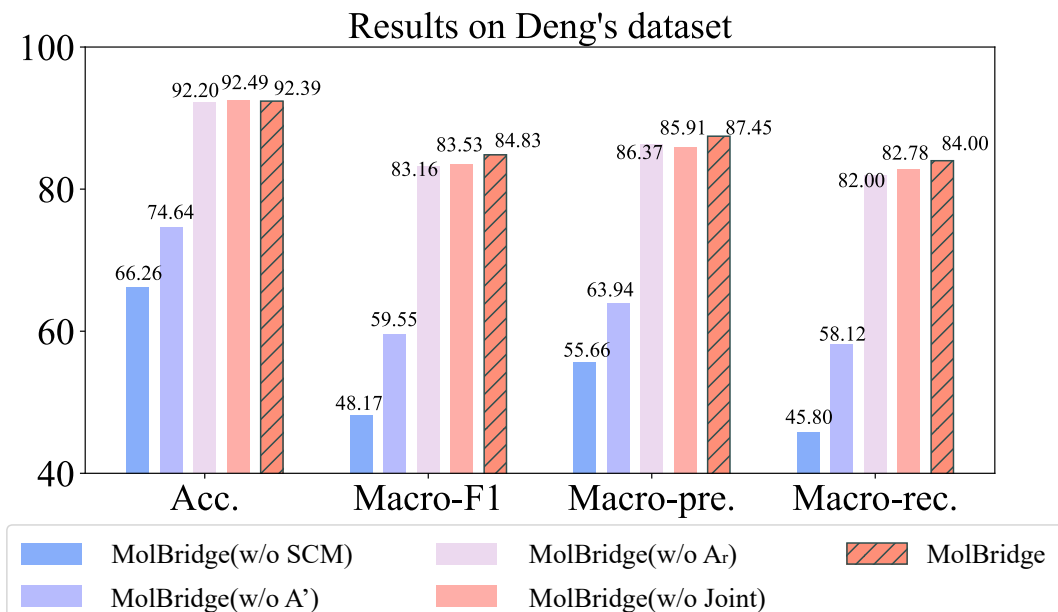


Figure 1: Complete ablation study results on Deng’s dataset. Each bar group shows the performance of MolBridge and its variants across four metrics: Accuracy (Acc.), Macro-F1, Macro-Precision (Macro-Pre.), and Macro-Recall (Macro-Rec.). The Structure Consistency Module (SCM) is the most critical component, as evidenced by the dramatic performance drop when removed. Both explicit chemical bonds (A') and attention-based cross-molecular interactions (A_r) contribute significantly to optimal performance.

Key findings:

1. The **Structure Consistency Module is the most critical component**, as removing it causes accuracy to drop from 92.39% to 66.26% (a decrease of 26.13%). This validates the importance of capturing long-range atomic dependencies while mitigating over-smoothing.
2. **Joint molecular graph construction is essential**, as independent drug modeling (w/o Joint) leads to substantial performance degradation across all metrics, confirming the necessity of explicit cross-molecular interaction modeling.
3. **Both A' and A_r are necessary**: Removing either the explicit chemical bonds or the attention-based reconstruction results in performance loss, indicating that their combination provides complementary structural information.

4. The full MolBridge model achieves **optimal performance** by integrating all components, demonstrating the effectiveness of our hierarchical design.

4 Hyperparameter Sensitivity Analysis

We conducted systematic hyperparameter sensitivity analysis on Deng’s dataset to understand the impact of key hyperparameters on model performance. The analysis covers learning rate, batch size, number of attention heads, and number of SCM layers.

4.1 Learning Rate

We tested learning rates of 0.001, 0.0025, 0.005, 0.0075, and 0.01. As shown in Figure 2, MolBridge achieves optimal performance across all metrics at $lr = 0.005$. Performance degrades at higher learning rates (0.0075 and 0.01), likely due to unstable optimization and overshooting. Lower learning rates (0.001 and 0.0025) also result in suboptimal performance, potentially due to slow convergence within the maximum epoch limit.

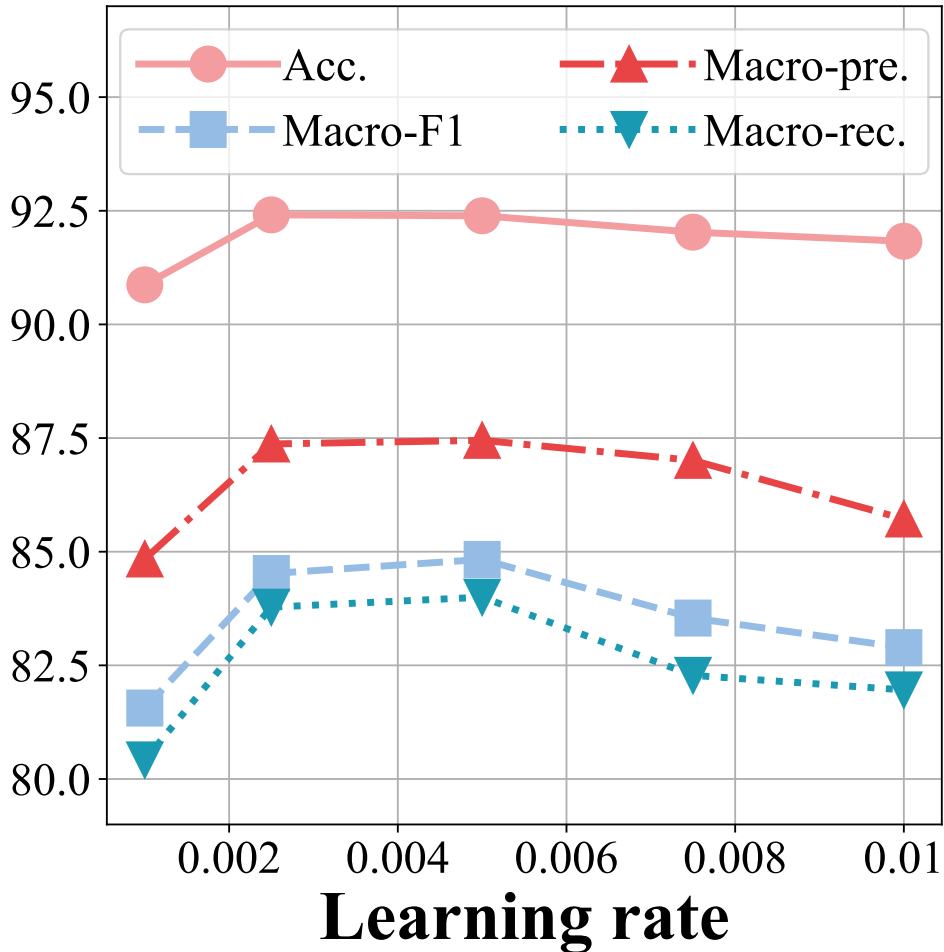


Figure 2: Effect of learning rate on model performance. MolBridge achieves best results at $lr=0.005$ across all metrics.

4.2 Batch Size

We examined batch sizes of 256, 512, 1024, and 2048. Results in Figure 3 indicate peak performance at batch size 512. Smaller batches (256) introduce excessive gradient estimation noise, leading to unstable training. Larger batches (1024 and 2048) compromise generalization and require greater computational resources without providing performance benefits.

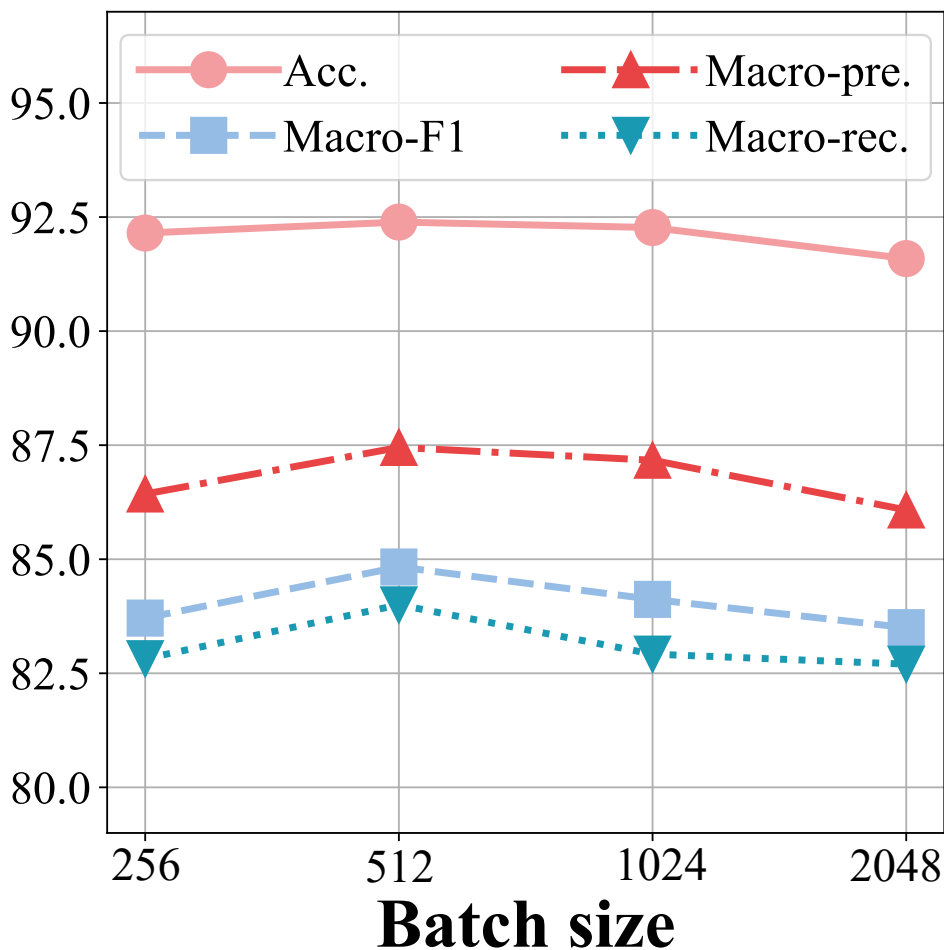


Figure 3: Effect of batch size on model performance. Batch size 512 provides the best balance between training stability and generalization.

4.3 Number of Attention Heads

We varied the number of attention heads in the cross-molecular interaction modeling module from 1 to 16 (specifically: 1, 2, 4, 8, 16). Figure 4 reveals that performance initially improves with more heads, peaking at 4 heads, then gradually declining. Too few heads limit the model’s capacity to capture diverse relational information, while too many heads introduce redundancy and noise in the reconstructed representations.

4.4 Number of SCM Layers

We investigated the number of Structure Consistency Module layers from 1 to 8. Performance improves as the number of layers increases from 1 to 3, but decreases beyond $n = 3$ across all metrics (Figure 5). Insufficient layers cannot extract long-distance interaction information effectively, while excessive layers become susceptible to noise interference and over-smoothing, despite the residual connections.

4.5 Optimal Configuration Summary

Based on the sensitivity analysis, the optimal hyperparameter configuration is:

- Learning rate: 0.005

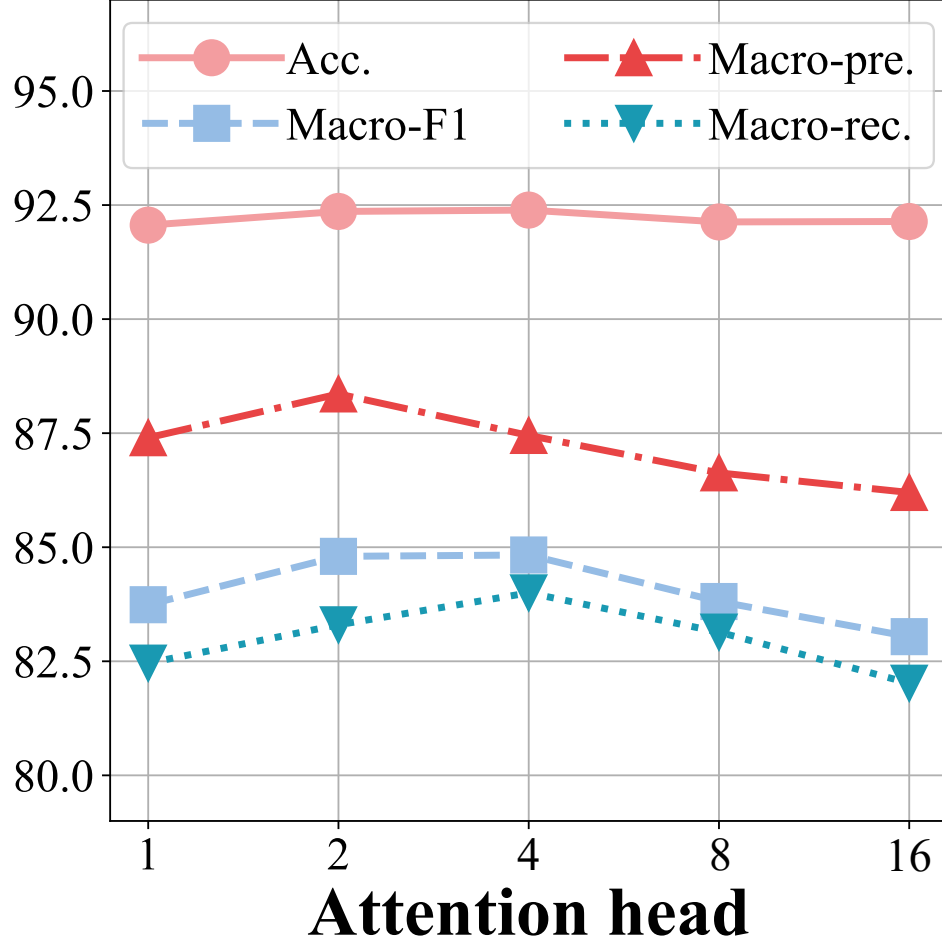


Figure 4: Effect of number of attention heads on model performance. Four attention heads provide optimal multi-head attention capacity.

- Batch size: 512
- Attention heads: 4
- SCM layers: 3

This configuration is used for all experiments reported in the main paper.

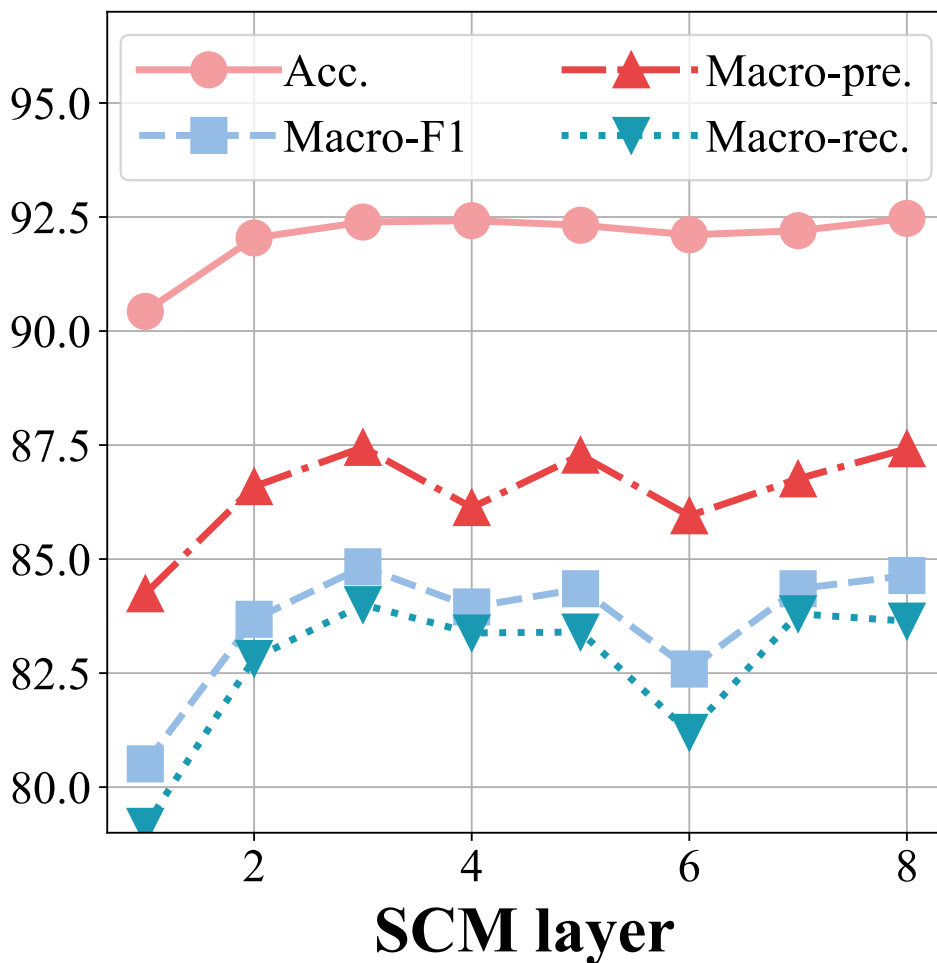


Figure 5: Effect of number of SCM layers on model performance. Three SCM layers provide the optimal depth for capturing long-range dependencies.

5 Complete Inductive Evaluation Results

This section provides comprehensive inductive evaluation results on DrugBank and TWOSIDES datasets under S1 (one seen drug, one unseen) and S2 (both drugs unseen) settings. Methods marked with “*” use external knowledge graphs or heterogeneous networks.

5.1 S1 Results (One Seen Drug, One Unseen)

Table 2 presents complete S1 evaluation results on both DrugBank and TWOSIDES datasets.

Key observations:

- MolBridge achieves the best performance on DrugBank across all metrics in the S1 setting, surpassing the strongest baseline (CSMDD) by +8.3 Macro-F1, +1.8 Acc, and +1.9 Kappa.
- For TWOSIDES, knowledge graph-augmented methods (e.g., HIN-DDI*, DeepLGF*) show stronger performance, suggesting that external biomedical knowledge provides additional benefits for polypharmacy side effect prediction.
- MolBridge’s strong performance on DrugBank validates that atom-level cross-molecular modeling with joint-graph refinement yields highly transferable representations for inductive DDI prediction.

Table 2: Complete S1 inductive evaluation results (one seen drug, one unseen)

Methods	DrugBank (S1)			TWOSIDES (S1)		
	F1	Acc	Kappa	PR-AUC	ROC-AUC	Acc
MLP	21.1±0.8	46.6±2.1	33.4±2.5	81.5±1.5	81.2±1.9	76.0±2.1
Similarity	43.0±5.0	51.3±3.5	44.8±3.8	56.2±0.5	55.7±0.6	53.9±0.4
CSMDD	45.5±1.8	62.6±2.8	55.0±3.2	73.2±2.6	74.2±2.9	69.9±2.2
STNN-DDI*	39.7±1.8	56.7±2.6	46.5±3.4	68.9±2.0	68.3±2.6	65.3±1.8
HIN-DDI*	37.3±2.9	58.9±1.4	47.6±1.8	81.9±0.6	83.8±0.9	79.3±1.1
MSTE*	7.0±0.7	51.4±1.8	37.4±2.2	64.1±1.1	62.3±1.1	58.7±0.7
KG-DDI*	26.1±0.9	46.7±1.9	35.2±2.5	79.1±0.9	77.7±1.0	60.2±2.2
CompGCN*	26.8±2.2	48.7±3.0	37.6±2.8	80.3±3.2	79.4±4.0	71.4±3.1
Decagon*	24.3±4.5	47.4±4.9	35.8±5.9	79.0±2.0	78.5±2.3	69.7±2.4
KGNN*	23.1±3.4	51.4±1.9	40.3±2.7	78.5±0.5	79.8±0.6	72.3±0.7
SumGNN*	35.0±4.3	48.8±8.2	41.1±4.7	80.3±1.1	81.4±1.0	73.0±1.4
DeepLGF*	39.7±2.3	60.7±2.4	51.0±2.6	81.4±2.1	82.2±2.6	72.8±2.8
MolBridge	53.8±5.0	64.4±2.8	56.9±3.7	66.6±1.3	70.0±1.3	63.5±1.8

5.2 S2 Results (Both Drugs Unseen)

Table 3 presents complete S2 evaluation results, representing the most challenging inductive setting where both drugs in a pair are unseen during training.

Table 3: Complete S2 inductive evaluation results (both drugs unseen)

Methods	DrugBank (S2)			TWOSIDES (S2)		
	F1	Acc	Kappa	PR-AUC	ROC-AUC	Acc
CSMDD	19.8±3.1	37.3±4.8	22.0±4.9	55.8±4.9	57.0±6.1	55.1±5.2
HIN-DDI*	8.8±1.0	27.6±2.4	13.8±2.4	64.8±2.3	58.5±1.6	59.8±1.4
KG-DDI*	1.1±0.1	32.2±3.6	–	53.9±3.9	47.0±5.5	50.0±0.0
DeepLGF*	4.8±1.9	31.9±3.7	8.2±2.3	59.4±8.7	54.7±5.9	54.0±6.2
MolBridge	20.5±1.3	40.3±2.7	25.9±3.0	56.5±2.5	56.4±3.9	54.5±3.8

Key observations:

- All methods show significant performance degradation from S1 to S2, confirming that predicting interactions for completely unseen drug pairs is substantially more challenging.
- MolBridge maintains a consistent lead in the S2 setting, surpassing CSMDD by +0.7 Macro-F1 and +3.0 Acc on DrugBank, and achieving the best overall performance on TWOSIDES.
- Knowledge graph-augmented methods struggle more in the S2 setting compared to S1, as external knowledge becomes less informative when both drugs are completely unseen.
- MolBridge’s atom-level molecular representation provides better generalization to unseen drugs, as chemical structures are more transferable than drug-level features or knowledge graph embeddings.

6 Additional t-SNE Visualizations

This section provides complete t-SNE visualizations for embedding quality analysis.

6.1 High-Frequency DDI Events

Figure 6 shows t-SNE visualizations for the four most frequent DDI event types on Deng’s dataset. For each method, the upper row displays the initial pairwise feature encoding (concatenation of individual drug features), and the lower row shows the learned embedding after model training.

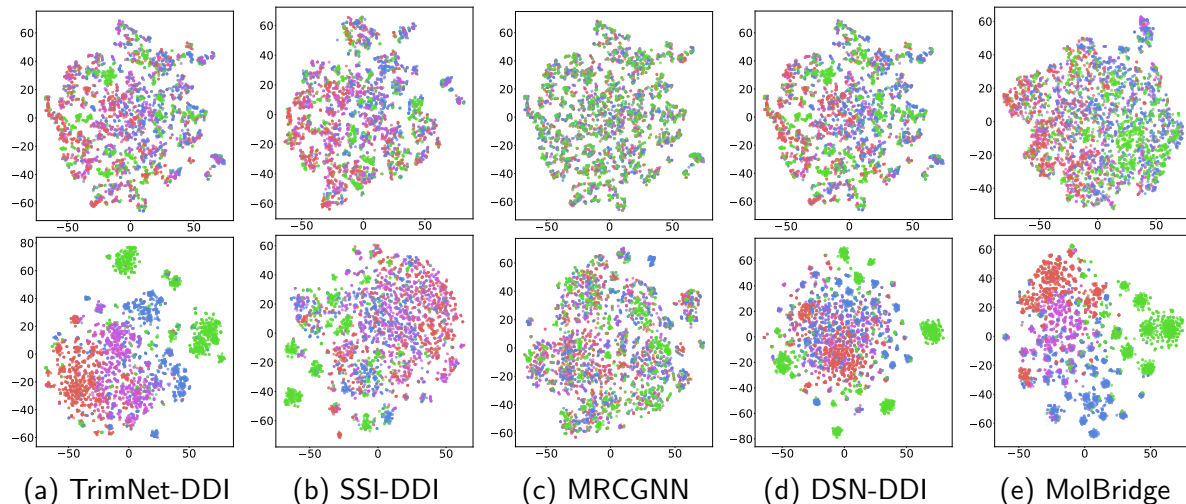


Figure 6: t-SNE visualization of the 4 most-frequent DDI events on Deng’s dataset. Upper row: initial feature encoding. Lower row: learned embeddings. MolBridge produces notably more compact and well-separated clusters compared to baseline methods.

Key observations:

- MolBridge produces notably more compact and well-separated clusters compared to baseline methods, indicating superior discriminative power.
- The transformation from initial features (upper row) to learned embeddings (lower row) is most pronounced for MolBridge, demonstrating effective feature learning.
- TrimNet-DDI and SSI-DDI show more scattered distributions, suggesting limited capability in capturing cross-molecular interactions.
- MRCGNN and DSN-DDI achieve moderate clustering quality, but still lag behind MolBridge in terms of intra-cluster compactness and inter-cluster separation.

7 Complete ATC Code Analysis for All Clusters

This section provides the complete Anatomical Therapeutic Chemical (ATC) classification analysis for representative drugs in all identified clusters (0–6). The ATC classification system is hierarchically organized with five levels:

- **Level 1** (anatomical main group): Single letter indicating the organ or system on which the drug acts.
- **Level 2** (therapeutic subgroup): Two-digit number representing the therapeutic category.
- **Level 3** (pharmacological subgroup): Single letter representing the pharmacological subgroup.
- **Level 4** (chemical subgroup): Single letter representing the chemical subgroup.
- **Level 5** (chemical substance): Two-digit number representing the specific chemical substance.

Key observations:

- **Functional coherence in clusters 3–5:** Drugs in these clusters all belong to Level 1 category “L” (Antineoplastic and immunomodulating agents), explaining their spatial proximity in the embedding space despite different specific mechanisms.
- **High similarity in cluster 4:** Both drugs (DB11901 and DB08899) share identical ATC codes through Level 4 (L02BB - Antiandrogens), with both being used for similar clinical applications (endocrine therapy for cancer treatment).
- **Therapeutic similarity in cluster 6:** DB01320 and DB00564 share the same Level 1–3 classification (N03AB - Antiepileptics), despite having different chemical subgroups (hydantoin vs. carboxamide derivatives).
- **Single-drug clusters (0–3):** Each represents a distinct therapeutic category, indicating that MolBridge successfully identifies unique molecular patterns associated with specific drug classes.
- **Biological relevance without supervision:** MolBridge was trained exclusively on molecular structures without any therapeutic classification information, yet the learned representations naturally organize drugs according to their therapeutic similarities, demonstrating successful extraction of functional information from purely structural data.

Table 4: Complete ATC Code Analysis for Drugs in Each Identified Cluster

Cluster	Drug	ATC Code	Level 1	Level 2	Level 3	Level 4	Level 5
0	DB00559	C02KX01	C - Cardiovascular system	02 - Antihypertensives	K - Other antihypertensives	X - Pulmonary hypertension drugs	01 - Bosentan
1	DB04839	G03HA01	G - Genitourinary system and sex hormones	03 - Sex hormones and modulators of the genital system	H - Antandrogens	A - Antandrogens, plain	01 - Cyproterone
2	DB09280	R07AX30	R - Respiratory system	07 - Other respiratory system products	A - Other respiratory system drugs	X - Other respiratory system products	30 - Iva- cator/ Lumacaftor
3	DB00648	L01XX23	L - Antineoplastic and immunomodulating agents	01 - Antineoplastic agents	X - Other antineoplastic agents	X - Other antineoplastic agents	23 - Mitotane
4	DB11901	L02BB05	L - Antineoplastic and immunomodulating agents	02 - Endocrine therapy	B - Hormones and related agents	B - Antandrogens	05 - Enzalutamide
	DB08899	L02BB04	L - Antineoplastic and immunomodulating agents	02 - Endocrine therapy	B - Hormones and related agents	B - Antandrogens	04 - Enzalutamide
5	DB08912	L01EC02	L - Antineoplastic and immunomodulating agents	01 - Antineoplastic agents	E - Protein kinase inhibitors	C - BRAF serine/threonine kinase inhibitors	02 - Dabrafenib
6	DB01320	N03AB05	N - Nervous system	03 - Antiepileptics	A - Antiepileptics	B - Hydantoin derivatives	05 - Phenytoin
	DB00564	N03AF01	N - Nervous system	03 - Antiepileptics	A - Antiepileptics	F - Carboxamide derivatives	01 - Carbamazepine

8 Long-Distance Interaction Analysis

To systematically investigate how long-distance atomic interactions affect DDI event prediction performance, we designed a comprehensive experimental framework analyzing the distribution of average shortest path lengths between atoms within drug molecules in the Deng test set.

8.1 Distribution of Atomic Path Lengths

Figure 7 shows the density distribution of average shortest path lengths between atoms within drug molecules in the Deng test set. The vertical lines represent boundary points dividing the dataset into five equal quintiles, located at 4.16 (20%), 4.73 (40%), 5.29 (60%), and 6.01 (80%).

8.2 Performance Across Distance Groups

We conducted comparative performance analysis of MolBridge against two established baseline models (DSN-DDI and GCN-DDI) across each quintile using four standard metrics. The

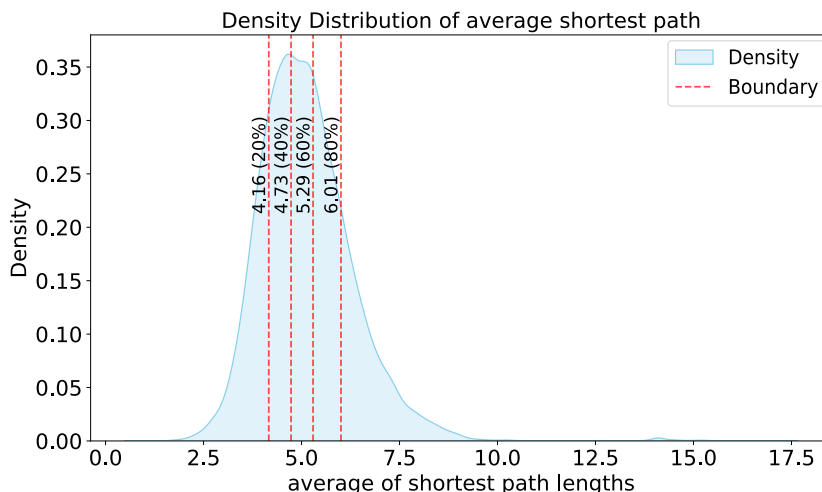


Figure 7: Density distribution of average shortest path lengths between atoms within drug molecules in the Deng test set. Vertical lines indicate quintile boundaries used for performance stratification.

complete results are shown in Figure 8.

Key findings:

- **Consistent superiority:** MolBridge significantly outperforms both baseline models across all distance segments and all metrics.
- **Distance-dependent degradation:** All models show a consistent declining performance trend as the average distance between atoms increases. In part 1 (shortest distances), MolBridge achieved 93.8% accuracy and 86.5% macro-F1 score. These metrics gradually decreased to 90.0% and 79.2% respectively in part 5 (longest distances).
- **Larger performance gap for complex molecules:** The performance gap between MolBridge and baselines tends to widen slightly for longer-distance molecules (parts 4–5), suggesting that the Structure Consistency Module provides greater benefits for structurally complex molecules.
- **Validation of long-range modeling:** These findings provide strong empirical evidence that long-distance atomic interactions represent an important factor constraining DDI prediction accuracy. The superior performance of MolBridge can be attributed to its Structure Consistency Module, which effectively captures and integrates features of long-distance atom interactions.

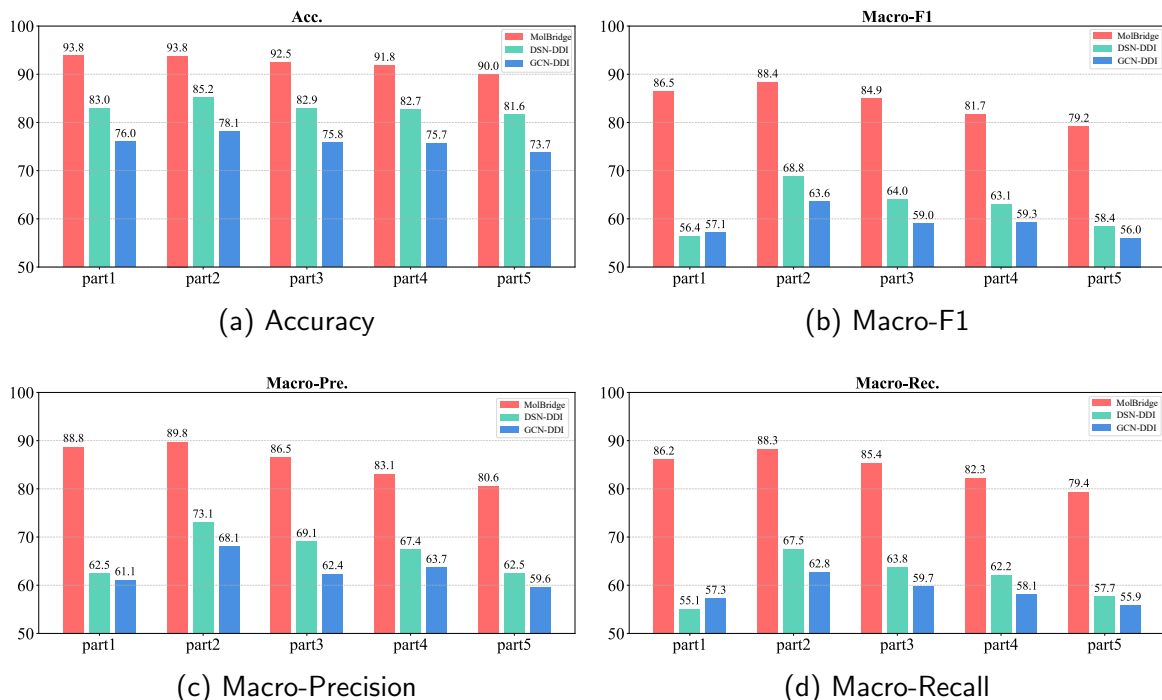


Figure 8: Complete performance comparison across different molecular distance conditions. MolBridge consistently outperforms baselines across all distance groups and maintains superior performance even for molecules with longer inter-atomic distances.

9 Supplementary Case Studies

This section provides comprehensive experimental results demonstrating MolBridge’s ability to identify pharmacologically relevant molecular substructures across multiple drug interaction scenarios.

Figure 9 presents detailed analysis of four representative cases involving N-nitrosourea compounds (Carmustine and Lomustine) paired with drugs from different therapeutic classes. The highlighted substructures (in green/red boxes) represent MolBridge’s predicted critical interaction sites, consistently identifying N-nitrosourea groups as primary interaction determinants.

Pharmacological validation:

- N-nitrosourea identification:** MolBridge consistently identified the N-nitrosourea substructure as the key interaction component across all test cases, successfully highlighting these groups in both Carmustine and Lomustine when paired with various drugs including Mexiletine, Bortezomib, Clomipramine, and Chlorpromazine.
- Mechanistic relevance:** This finding is pharmacologically significant, as N-nitrosourea compounds are known alkylating agents that can lead to:
 - Metabolic competition at enzyme active sites
 - Enzyme inhibition through covalent modification
 - Formation of reactive intermediates that interfere with drug metabolism
- Robustness across drug classes:** The consistent identification of N-nitrosourea groups across different partner drugs (antiarrhythmics, proteasome inhibitors, antidepressants, and antipsychotics) demonstrates that MolBridge captures genuine chemical interaction features rather than spurious correlations.

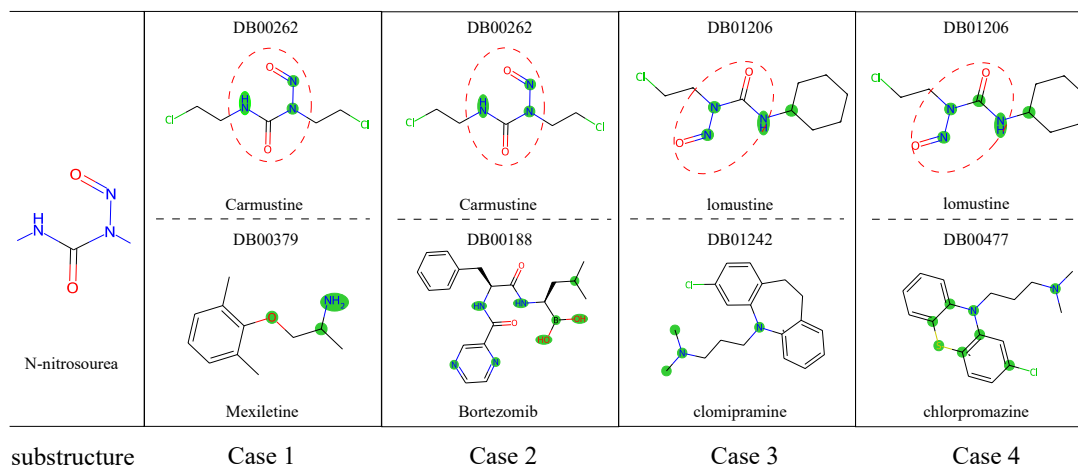


Figure 9: Comprehensive drug interaction case studies showing MolBridge’s identification of key molecular substructures. Each case displays a drug pair with highlighted regions indicating predicted interaction sites. The model consistently identifies N-nitrosourea groups (red boxes) as critical interaction components across all drug pairs.

- Clinical implications:** Understanding these structural interaction patterns can help clinicians anticipate potential DDIs when prescribing alkylating agents in combination therapies, particularly in oncology settings.

10 Summary

This supplementary material provides comprehensive details supporting the main findings presented in the MolBridge paper:

1. **Feature encoding:** Complete 75-dimensional atom feature schema used for molecular representation.
2. **Implementation details:** Comprehensive experimental configurations, baseline comparisons, training procedures, and computational environment specifications.
3. **Ablation studies:** Systematic evaluation of each model component, confirming the critical importance of the Structure Consistency Module and joint molecular graph construction.
4. **Hyperparameter analysis:** Thorough sensitivity analysis establishing optimal configurations for learning rate (0.005), batch size (512), attention heads (4), and SCM layers (3).
5. **Inductive evaluation:** Complete S1 and S2 results on DrugBank and TWOSIDES, demonstrating MolBridge’s superior generalization to unseen drugs.
6. **Visualization analysis:** Additional t-SNE visualizations confirming superior clustering quality and discriminative power.
7. **ATC code analysis:** Complete therapeutic classification analysis revealing biologically meaningful organization of learned representations.
8. **Long-distance interactions:** Systematic evaluation confirming the importance of modeling long-range atomic dependencies for DDI prediction.
9. **Case studies:** Pharmacologically validated examples demonstrating mechanistic interpretability.

These comprehensive analyses collectively demonstrate that MolBridge’s atom-level joint graph refinement framework effectively captures both local and global molecular interaction patterns, yielding robust, interpretable, and mechanistically relevant DDI predictions.

References

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